

BrainRobotics Hand

User Manual

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1. Product general description

BrainRobotics hand is an intelligent product that highly integrates brain-computer interface technology and artificial intelligence algorithms. This bionic hand can recognize the wearer's movement intention by detecting the wearer's neural and muscular electrical signals, and convert the intention into the action of the bionic hand, so as to move the hand freely.

BrainRobotics hand has 11 movable joints and 6 driving degrees of freedom, which can realize the independent movement of 5 fingers, wrist, and the cooperative operation between fingers. Furthermore, it will meet the requirement of common gestures in the daily use of upper limb amputees, and help them controlling the bionic hand as flexibly as controlling their own hands.



2. Key features



3. Awards and prizes



《Time》
The 100 best inventions of 2019



CES 2019
Innovation Awards



Red Dot Winner 2020
Best of the best



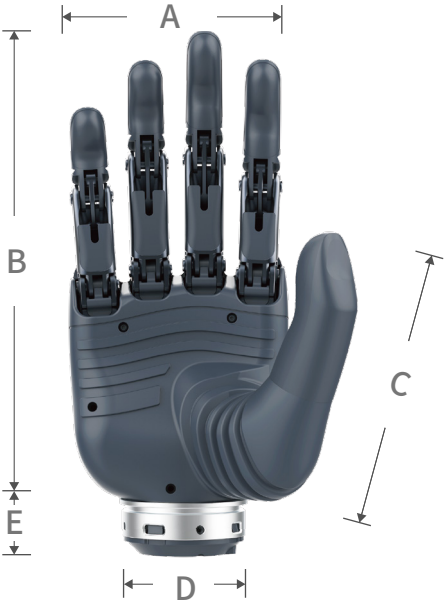
CES 2020
TOP TECH WINNER

4. Editions of BrainRobotics hand

	Premium edition	Standard edition
Controlling algorithm	Advanced phantom-limb motor-imagery control (based on brain-computer interface technology and artificial intelligence algorithm)	Multi-gesture motion control (based on high-precision EMG and intelligent signal processing algorithm)
Operating mode	Basic mode (preset gesture switching) Advanced mode (single finger control, no switching, unlimited gestures)	Basic mode (preset gesture switching)
Intelligent hardware	Raspberry Pi Compute Module 3+	
Electrodes	8-channel electrodes	
Adaptation symptom	Wrist disarticulation, forearm amputation	Wrist disarticulation, forearm and upper arm amputation

5. Dimensions and parameters

Product Parameters	Medium hand	Medium hand	Mini hand	Mini hand
	Premium Edition	Standard Edition	Premium Edition	Standard Edition
A: Maximum palm width	86mm / 3.38inch		77mm / 3.03inch	
B: Length from middle finger tip to palm base	183mm / 7.2inch		163mm / 6.42inch	
C: Length from thumb tip to palm base	100mm / 3.94inch		95mm / 3.74inch	
D: Wrist Diameter	48mm / 1.89inch			
E: Wrist Height	20mm / 0.79inch			
Weight	546g		512g	
Color	Grey		Skin color	



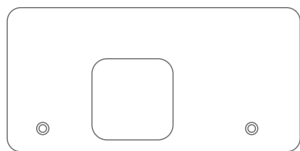
6. Technical information

Parameters		Data
Operating Parameters	Supply voltage	8.4V
	Maximum current	3A
	Battery capacity	2200mAh
	Battery endurance	Open and close fingers for 4,000 times
	Maximum opening distance (between index finger and thumb)	113mm / 4.45inch
	Maximum time for single finger bending and extension	0.8s
	Maximum fingertip pinch force (Single fingertip pinch force * 4)	120N
	Maximum fingertip self-locking force (Single fingertip self-locking force * 4)	>320N
	Thumb adduction and abduction range	90°
	Flexion and extension range of the thumb metacarpophalangeal joint	50°
	Flexion and extension range of the four finger metacarpophalangeal joints	90°
	Working temperature	-10°C ~ 45°C
	Fingers service life	Open and close fingers for 300,000 times
Hardware Components	Wireless communication	BLE
	Sensor setting	Position sensor, current sensor, gyroscope, accelerometer
	Mainbody material	Aerospace-grade aluminum alloy and food-grade rubber
	Mobility	11 freely movable joints, autonomous bi-axial thumb
Intelligent Software and System	Perception	Force, position, posture, vibration, perception
	Controlling	Movement intelligent control, fine motor skill control, anti-blocking control, anti-fall and shake protection
	Algorithm	AI algorithm based on deep learning

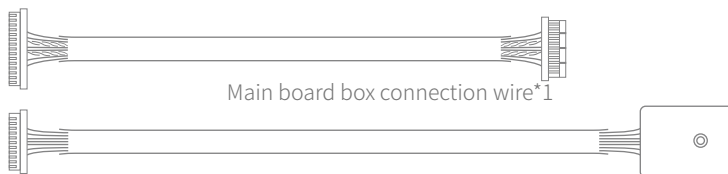
7. Accessory list

The accessory list of premium-edition bionic hand:

Accessory	Number	Function
Charger	1	Charge the battery through the port in the wrist.
Charger cable	1	Connect bionic hand with the charger and be used as a data cable for recording data.
Battery	1	Power the bionic hand with the port in the wrist.
Battery cable	1	Extend the battery wire.
Adaptor board	1	
Heat shrink tube	1	Fix the battery cable and adapter board.
Main board box	1	Connect the electrodes with bionic hand to collecting and processing signals.
Main board box connection wire	1	Connect the main board box with the bionic hand. (already installed in the main board box)
Electrodes	8	Close to skin, sense the user's physiological electrical signals.
Electrode connection wires	2	Connect the electrodes with the main board box. (already installed in the main board box)
Electrode screws	16	Fix the electrodes on the socket.
Switch	1	Control bionic hand on and off.
Switch connection wire	1	Connect the switch to the bionic hand.
Switch screws and gaskets	2	Fix the power switch.
Hex wrench	1	Assemble bionic hand and accessories.
Wrist screws	4	Connect the socket with the bionic hand.

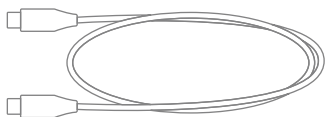


Main board box*1



Main board box connection wire*1

Electrode connection wires*2



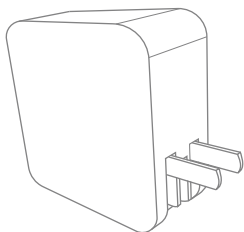
Charger cable*1



Electrodes*8



Electrode screws*16



Charger*1



Hex wrench*1



Battery cable*1



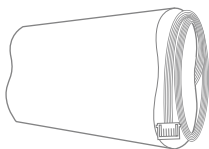
Wrist screws*4



Switch connection wire*1



Switch screws*2



Battery*1



Switch*1



Adaptor board*1



Heat shrink tube*1

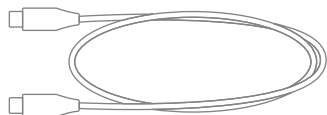


Gaskets*2

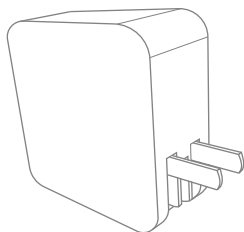
7. Accessory list

The accessory list of standard-edition bionic hand:

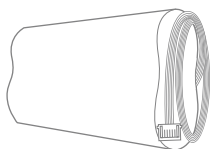
Accessory	Number	Function
Charger	1	Charge the battery through the port in the wrist.
Charger cable	1	Connect bionic hand with the charger and be used as a data cable for recording data.
Battery	1	Power the bionic hand with the port in the wrist.
Battery cable	1	Extend the battery wire.
Adaptor board	1	
Heat shrink tube	1	Fix the battery cable, battery wire and adapter board.
Electrode connector	1	Connect the electrodes with bionic hand to collecting signals.
Electrode wire	1	Connect the electrode connector with the bionic hand.
Electrode	2	Close to skin, sense the user's physiological electrical signals.
Electrode screws	4	Fix the electrodes on the socket.
Switch	1	Control bionic hand on and off.
Switch connection wire	1	Connect the switch to the bionic hand.
Switch screws and gaskets	2	Fix the power switch.
Hex wrench	1	Assemble bionic hand and accessories.
Wrist screws	4	Connect the socket with the bionic hand.



Charger cable*1



Charger*1



Battery*1



Hex wrench*1



Switch*1



Adaptor board*1



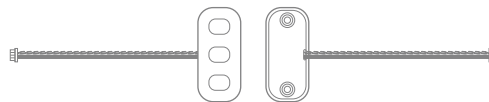
Electrode connector*1



Heat shrink tube*1



Gaskets*2



Electrodes*2



Electrode screws*4



Battery cable*1



Wrist screws*4



Switch connection wire*1



Electrode wire*1

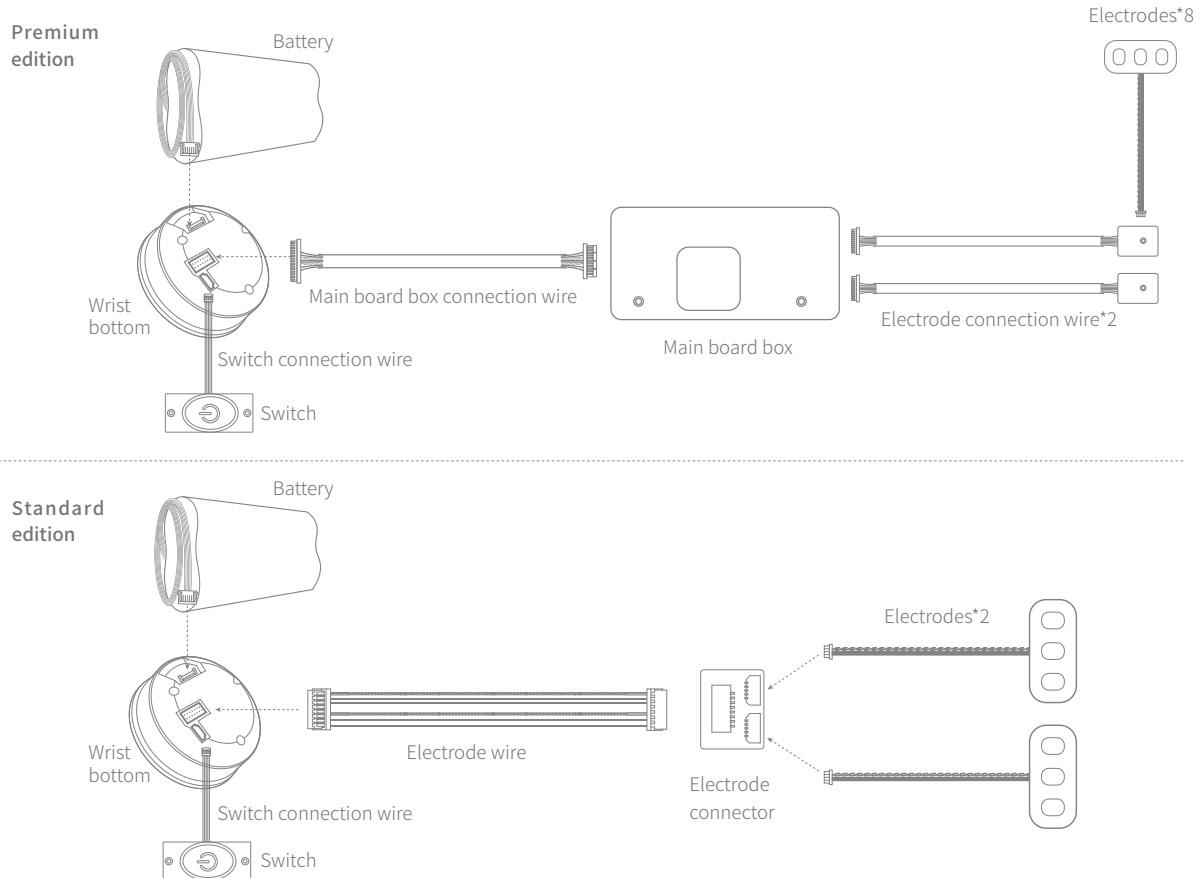


Switch screws*2

8. Bionic hand assembly process

a Wire connection

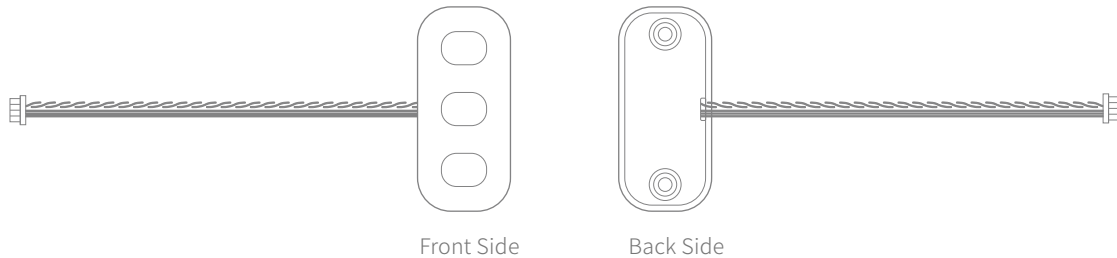
Before using for the first time, please ask the assembler to connect the bionic hand and accessories as shown in the following figure, and put the battery, connecting wire, electrode, etc. into the socket.



b Connection of wrist and socket

The wrist is securely fixed to the socket through 4 wrist mounting holes. Installation can be done using the wrist screws and a hex wrench that are provided with the bionic hand. Before installing the wrist, it is necessary to complete the assembly and connection of other parts with the bionic hand according to the illustration in 8.a.

c Electrodes installation

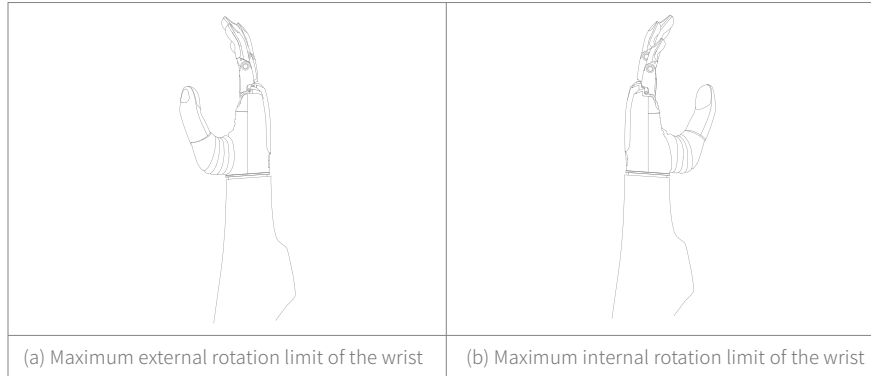


In the process of making the socket, it should be ensured that the front side of the electrode has good contact with the skin, the position is correct, and the signal quality meets the requirements. This electrode product does not require adjustment of magnification. After installing the electrode into the socket, wire the electrode as shown in 8.a.

9. Key points of usage

a Wrist

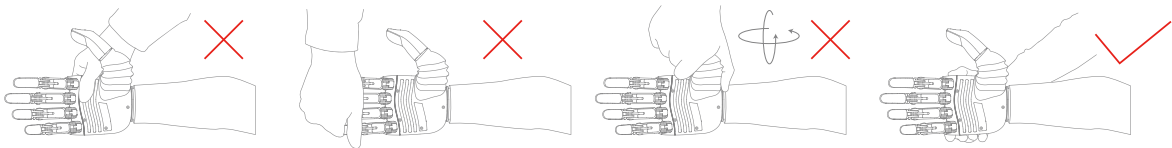
The wrist of BrainRobotics Hand has a passive internal-external rotation range of 180° . The rotation range is depicted in the figure below. The starting position of the rotation range is determined by the position of the mounting hole on the socket. The wrist has a total of 7 locking positions, with a 30° interval between adjacent locking positions.



To rotate the wrist, users should grip the hard shell of the palm. To protect your bionic hand, please take note of the following when rotating the wrist:

- Avoid applying rotational force on the finger components or soft parts.
- Avoid keeping the hand (wrist) in a non-locking position between adjacent locking positions for long terms.
- Avoid forcefully rotating the wrist beyond the rotational range at the starting or ending positions.
- Avoid frequent and rapid rotations of the wrist.

The figure below illustrates the correct and incorrect ways to operate the wrist.



b Indicator and switch

Press the switch button located in the socket to start up or shut down. In the "on" state, the hand is powered on and works normally. In the "off" state, the hand is powered off and stops working.

An indicator on the back of the bionic hand shows the charge. When the indicator blinks red, it indicates that the battery capacity is less than 10%. Please charge it in time. In addition, the indicator is a button whose functions includes unlocking the wrist and switching gestures (for standard edition).

c Charging instructions

△When charging, please follow the sequence below strictly:

1. Turn off the power source of the hand;
2. Remove the hand from the body;
3. Confirm once again that the power of the hand is turned off;
4. Insert the charger cable's connector into the charging port of the hand;
5. Plug the AC plug of the charger into the mains outlet.

△ For user safety, please remove the bionic hand from the body and turn the power switch to the "OFF" position before charging. It is necessary to use the provided charger to charge the hand.

△ When charging the battery, please place the hand flat and avoid any manipulation or operation of the hand.

△ A fully discharged battery typically takes approximately 3 hours to fully charge. Therefore, it is recommended to charge the bionic hand overnight when it is not in use to ensure its normal operation.

d Charging indicator

The charging indicator light is located on the wrist of the bionic hand, and its indicator light color represents the following meanings:

Steady red light		Charging
Steady green light		Fully charged
Alternating flashing of red and green lights	 	Error occurred

△If a charging error occurs, please follow the steps below:

1. Check if the power is turned off. If it is not, turn the switch to the "OFF" position.
2. Ensure that the provided BrainCo charger is being used. If not, replace it with the appropriate charger.
3. Disconnect and then reconnect the charging port and the cable.
4. Check the environment temperature to ensure it is within the range of 10°C to 45°C. Avoid extreme temperatures that are too high or too low.

If, after these steps, the charging indicator light continues to flash alternately between red and green, please contact the after-sales support personnel for further assistance.

Other consideration while charging:

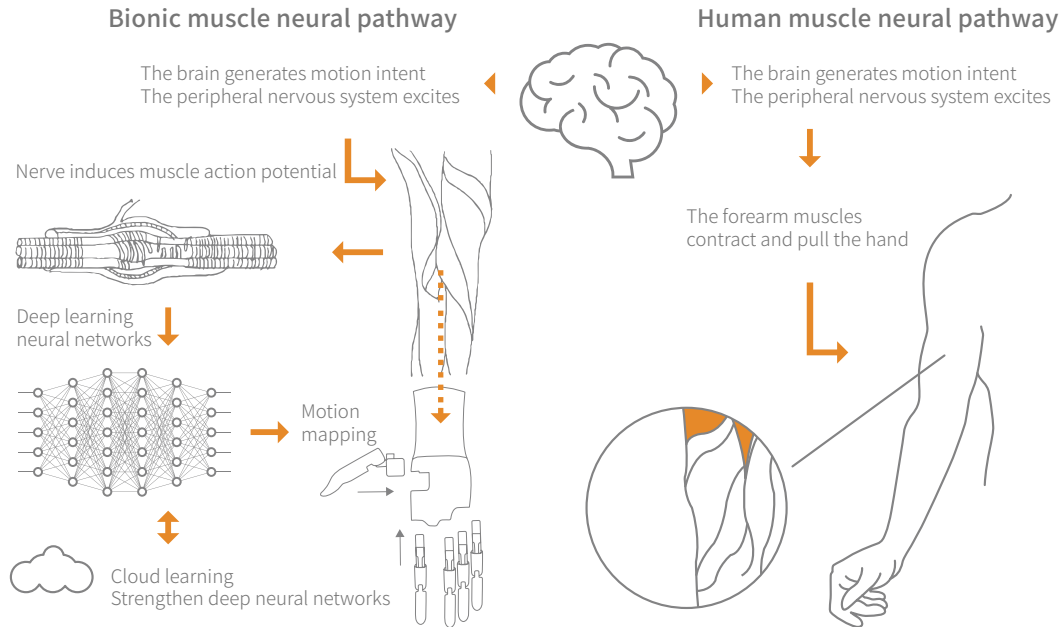
1. The charging environment temperature should be within the range of 10°C to 45°C. If the temperature exceeds this range, the charging time may need to be extended or charging may not be possible.
2. To determine the completion of charging, please rely on the green light indication of the charging indicator.
3. Do not pull on the cables to remove the charger. Please hold the charger to charge or stop charging.

10. Functional theory and controlling principle

a. Premium edition

The premium edition of BrainRobotics hand uses advanced phantom-limb motor-imagery control technology, based on brain-computer interface technology and artificial intelligence algorithms, to help the wearer convert their intention into the action of the bionic hand. At the same time, according to different application scenarios and usage habits, it provides two operation modes of "basic" and "advanced", which is convenient for the wearer to choose flexibly.

The basic principle of the premium edition bionic hand is shown below: When the wearer tries to make an action, the intention to move is generated in the brain, and the nerve in the arm is induced to generate an action potential, causing the muscles to respond. The electrodes in the socket pick up the corresponding changes and convert them into electrical signals that are transmitted to the bionic hand. After processing by the deep learning neural network, the relevant information will be mapped to each joint of the bionic hand to complete the action. In addition, the bionic hand will also enhance the deep neural network through cloud learning to achieve higher recognition accuracy.



The premium BrainRobotics hand can switch operation modes on the app. Both modes require training the model by moving the phantom limb, which allows the bionic hand to identify the wearer's intentions more accurately. When the app side selects to start recording gestures, the wearer imagines the phantom limb movement, and the EMG data set will be passed into the model. For specific training methods and operations, see the tutorial on the app.

When adding a new model on the app side, select "Basic Mode" : the wearer can select up to 6 common gestures including "relax", "clench" and "open" among many preset gestures to train the existing model. When the similarity of the trained model reaches a certain requirement, the bionic hand can make corresponding preset gestures as long as the phantom-limb motor-imagery is performed.

When adding a new model on the app side, select "Advanced Mode" : in addition to recording the three must-play gestures of "relax", "clench" and "open", the wearer can also record the five-finger flexion and extension gestures respectively to conduct targeted training on the existing model. When the similarity of the trained model reaches a certain requirement, as long as the phantom-limb motor-imagery is performed, the bionic hand can make the desired gesture, which truly achieve dexterity, intelligence and freely move.

b. Standard edition

The standard edition of BrainRobotics hand is based on high-precision EMG and intelligent signal processing algorithms, which can control a variety of gesture movements. In the app, there are a total of 24 discrete gestures, and the wearer can choose up to six of them to be attached to the bionic hand at the same time. The wearer can switch discrete gestures by controlling the times and durance of wrist flexion and extension force. Take the "five-finger grip" gesture as an example, control the wrist flexion and extension with short-term force once, the indicator light will be steady red, and the bionic hand will make the "five-finger grip" gesture. In addition, the wearer can also switch the preset action by tapping the indicator button. If the "five-finger grip" is set as the No. 1 action in the app, then press the back button of the hand once to make the "five-finger grip"; the No. 2 action needs to be pressed twice and the rest is the same.

Power
grip



Precision
pinch
grip



Tripod
pinch
grip



Index
point



Lateral
grip



Hook
grip

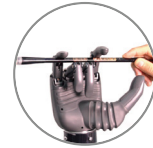




Finger point
(opposed thumb)



Tripod pinch



Pen-grip 1



Two-finger spread
(opposed)



Inward thumb



Pen-grip 2



Two-finger spread
(unopposed)



Flick
(index finger)



Cylindrical
grip



Precision pinch



Flick
(middle finger)



Shaka sign



Mouse
control



Four finger
straight



Hook



Rock and roll



Palm open



Pen-grip 3

11. Instructions for use

Bionic hand

1. The bionic hand is intended for personal use by the user only. It is strictly prohibited for others to operate the hand.
2. The bionic hand is not capable of sensing temperature and humidity. It should not be exposed to high temperatures or high humidity environments.
3. Avoid subjecting the bionic hand to excessive loads or impacts.
4. Unauthorized disassembly of the bionic hand will void the warranty as stated in the "After-Sales Policy." In case of any malfunctions, please contact the designated after-sales service provider.
5. Do not use the bionic hand in strong electromagnetic fields, such as near high-voltage power lines or high-power machinery.
6. Avoid vigorous shaking of the bionic hand.
7. Do not forcefully pull or tug the fingers of the bionic hand.
8. Maintenance and servicing of the bionic hand should not be performed while it is powered on.
9. Do not use the bionic hand to operate dangerous machinery. We are not liable for any personal injury or property damage caused by such actions.
10. Avoid applying excessive force to the fingertips of the bionic hand.
11. Without proper protection, do not allow the bionic hand to come into contact with liquids (alcohol, water, beverages, etc.) and dust. In case of accidental contact, immediately turn off the hand and contact the after-sales service provider.
12. Do not use the bionic hand to grasp objects that are too heavy, too hot, sharp, rough, or corrosive.
13. Keep the bionic hand away from open flames.
14. Do not place the bionic hand in flammable or explosive environments.
15. Do not charge the bionic hand while it is powered on.
16. Adhere to local laws and regulations when operating everyday machinery (e.g., driving a motor vehicle, operating machine tools) with the bionic hand. We are not liable for any personal injury or proper.

Battery

1. Cutting or modifying the battery wires is strictly prohibited.
2. Users are not allowed to replace the battery themselves.
3. Avoid bending or squeezing the battery.
4. Do not puncture the battery.
5. Do not dismantle the battery.
6. Keep the battery away from high-temperature environments.
7. Do not incinerate the battery.
8. Do not exchange battery interface cables.
9. Avoid short-circuiting the battery.
10. Do not dispose of the battery arbitrarily.

12. After-sales policies

The bionic hand as a whole (excluding the battery, silicone palm cover, and plastic fingers) is covered by a 3-year warranty from the date of purchase.;

- The battery is covered by a 1-year warranty from the date of purchase;
- The silicone palm cover is covered by a 1-year warranty, and the plastic fingers are also covered by a 1-year warranty;
- During the warranty period, one free professional maintenance service is provided annually. During the maintenance service, the silicone palm cover and plastic fingers will be replaced free of charge, and the gearboxes and motors will be inspected and serviced;
- Normal surface wear that does not affect the functioning is not covered by the warranty. For issues unrelated to product quality, no returns or exchanges are accepted;
- If any malfunctions occur with the bionic hand, please contact the designated bionic hand maintenance organization through your assembly institution.

The specific after-sales policy is subject to the commitments made by the distributor.

13. Disclaimer

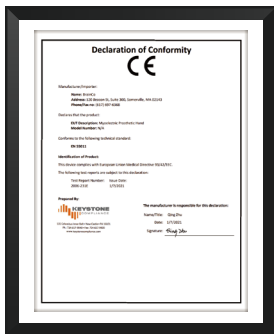
1. To protect the legitimate rights and interests of users, please carefully read the user manual provided with this product before using it. We reserve the right to update the aforementioned document.
2. By using this product, you are deemed to have read, understood, acknowledged, and accepted all the terms and contents of the user manual of this product. The user is responsible for their actions and all consequences arising therefrom. The user agrees to use this product solely for legitimate purposes and agrees to comply with the terms herein and any other policies or guidelines that may be established by us.
3. During the use of this product, it is imperative that you strictly comply with and adhere to the requirements outlined in the user manual , including but not limited to those related to safety. Any personal injury, accidents, property damage, legal disputes, and any other adverse events resulting from violations of the safety instructions or unforeseeable factors shall be the sole responsibility of the user, and our company shall not be liable for any damages.
4. Our company shall not be liable for any actions by the user that violate legal regulations, whether directly or indirectly arising from the use of this product.
5. Other disclaimers as provided by laws and regulations.

Note:

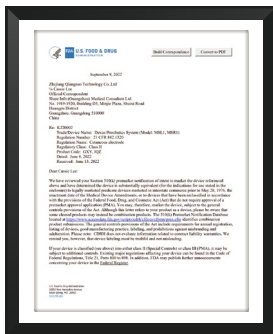
- (1) If the user is a minor under the age of 18, they should read this agreement carefully under the supervision and guidance of a guardian, and obtain the consent of the guardian before using this product.
- (2) Both the minor user and their guardian understand that if the minor violates laws, regulations, or the terms of this agreement, they will be held fully responsible for any legal liabilities that may arise, in accordance with applicable laws and regulations.

14. Certificates

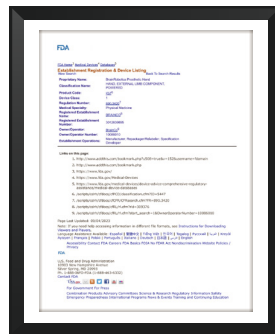
EU Declaration of Conformity
Test report number: 2006-231E



FDA510(k) Clearance
510(k) number: K220002



FDA Establishment Registration
Regulation number: 890.3420



15. Core patents



ZL201821201287.4



ZL202030054968.9



ZL202111337725.6



ZL202111482265.6



ZL202111562781.X



ZL202111562493.4



ZL202111576933.7



ZL202220259055.4



ZL202210132466.1



ZL202210575280.3



Customer service number: 400 6886 289